AZURE DAY 2026 HACKATHON · DISASTER RECOVERY AGENT

Disaster Recovery (DR) Agent

Autonomous Disaster Recovery Validation

Built 100% on Azure. AI agents that validate DR by executing real tests: fail over, measure RTO/RPO, fail back, and prove you can recover, on any day instead of once a year.

Fail over **Measure** **Fail back** **Prove**

Team Azure Armada · Virtusa

Downtime now costs about $900,000 an hour

The latest independent research (2026) and a 2025 cloud outage: the cost of not recovering on time.

$15,000

per minute of downtime

is the average for Global 2000 companies: about $900,000 an hour and $600B a year in total. Regulatory fines alone average $51M per company a year.

Splunk & Oxford Economics, The Hidden Costs of Downtime, May 2026 (2,000 executives) [1]

57%

had a last major outage over $100K

and 1 in 5 operators said their last major outage cost more than $1 million, for the second year running.

Uptime Institute, Annual Outage Analysis 2026 [2]

$500–650M

from one cloud-region outage

lost by US companies when AWS us-east-1 failed on 20 October 2025. Even hyperscalers fail; what matters is whether your recovery works.

Parametrix estimate, Oct 2025 [3]

At $15,000 a minute, one hour of downtime costs about $900,000. Recovery that is assumed, not proven, is an unpriced risk.

In India, failing to recover is punished by freezing growth

Regulators act on outages and weak disaster recovery, not just on breaches.

HDFC Bank — Dec 2020

After repeated outages, the latest caused by a power failure at a primary data centre, the RBI stopped all new digital launches and new credit-card sourcing.

≈15 months

until digital launches were allowed again (Mar 2022); cards resumed Aug 2021

[4]

NSE — Feb 2021

Trading was halted for nearly four hours on 24 February 2021. NSE and NSE Clearing settled with SEBI; executives were ordered to do community service and training.

≈ $8.8M

paid by NSE and NSE Clearing to settle with SEBI (2023)

[5]

Kotak Mahindra Bank — Apr 2024

The RBI cited deficiencies including business continuity and disaster recovery, and frequent outages, and barred online customer onboarding and new credit cards.

≈10 months

until the restrictions were lifted (Feb 2025)

[6]

For a bank, a 10–15 month ban on new digital customers costs far more than any fine.

Regulators now require proof of recovery, on a schedule

Having a DR plan is not enough. It must be tested, the gaps fixed, and the test repeated.

Regulator	What it requires	If you fall short
RBI **Cybersecurity, Technology: Risk, Resilience and Assurance Framework Directions, 31 Jul 2026**	DR drills for critical information systems at least half-yearly. Any major issue found must be fixed and re-tested successfully before the next cycle. [7]	Supervisory restrictions on business, as with HDFC Bank and Kotak Mahindra Bank. [4][6]
SEBI **Cybersecurity & Cyber Resilience Framework**	Resume critical operations within 2 hours (RTO) with a 15-minute RPO. Market infrastructure: run from the DR site within 45 minutes of declaring a disaster, drills every quarter, unannounced live trading from the DR site every six months. [8][9]	Enforcement and settlements, as with NSE: about $8.8M. [5]
EU DORA **Regulation (EU) 2022/2554, applies from 17 Jan 2025**	Test ICT business continuity and response and recovery plans at least yearly and after major changes (Art. 11(6)); test backup restoration periodically (Art. 12). [10]	Penalties set by each EU state (Art. 50). Critical ICT providers: up to 1% of average daily worldwide turnover, every day, for up to six months (Art. 35). [10]

Our demo uses the same yardstick: target RTO 30 min and RPO 15 min, stricter than SEBI's 2 hours, and every drill produces timestamped, audit-ready evidence.

Yet most organisations cannot prove their recovery works today

Plans exist on paper. Tests are rare, manual and partial, so the gaps surface during the real disaster.

< 1 in 3

organisations perform regular failover testing

Cockroach Labs, State of Resilience 2025 (1,000 executives, 2024 survey) [11]

20%

of executives believe their organisation is fully prepared to prevent or respond to outages

Cockroach Labs, State of Resilience 2025 [11]

#1

cause of human-error outages: staff not following established procedures

Uptime Institute, Annual Outage Analysis 2026 [2]

Why the gap persists

Tests are disruptive and manual

A real failover needs several teams, a change window and hand-written timings, so it happens rarely.

Environments drift between tests

A setting added in one region but not the other only shows up when you fail over.

Runbooks go stale

Documents name servers and services that no longer exist; engineers find out mid-incident.

RTO and RPO are assumed

Targets are signed off on paper; actual recovery time and data loss are rarely measured end to end.

What an unproven DR plan costs: one bad morning

Illustration for a tier-1 payments service, using the cited figures. Not a forecast; the inputs are shown.

Recovery target (RTO)	**30 minutes**	Signed off on paper; never measured end to end
Actual recovery	**4 hours**	Comparable to the NSE trading halt of Feb 2021 [5]
Hourly cost of downtime	**$900,000**	$15,000 a minute, the Global 2000 average (Splunk & Oxford Economics, 2026) [1]
Direct loss	**$3.6M**	4 h × $900K; $3.15M of it beyond the 30-minute target
Regulatory precedent	**≈ $8.8M**	What NSE paid to settle its 2021 outage case [5]; banks have faced 10–15 month business restrictions [4][6]

The DR Agent measured 29.04 s to recover the same kind of service in a live Azure drill (19 Oct 2026), and proved the data loss write by write: 2.8 s of writes, against a 15-minute target.

The DR Agent runs the real drill and proves the result

Six AI agents on Azure AI Foundry plan and judge; a guarded engine executes on live Azure resources; humans approve.

Phase 1

Design & prepare

Architect	picks the cheapest strategy that truly meets RTO/RPO, with real Azure prices
Readiness	7 live checks; finds drift before the disaster; one-click approved fix
Runbook	flags stale steps and writes the corrected runbook

Phase 2

Plan, execute, fail back, report

Planner	builds the drill from an allowlist of safe actions; the engine rejects any unsafe plan
Engine + Validator	injects the outage, fails over, measures; diagnoses any failure live
Reporter	audit-ready report with every second of RTO accounted for

	Today	With the DR Agent
How often DR is proven	Once or twice a year	On demand, any day
RTO and RPO	Assumed on paper	Measured on every drill, to the millisecond
Configuration drift	Found during the real outage	Found by Autopilot every 10 min, fixed with approval
Audit evidence	Screenshots and spreadsheets	Generated report with timestamps and approvals

Six specialist agents, each with one job and a contract

Versioned prompt agents in Azure AI Foundry Agent Service. Every answer must match a strict JSON schema; every decision that drives an action is checked by code.

Agent	Model	Gets (facts from code)	Returns	Checked by code
Architect	`gpt-4.1-mini`	Profile, inventory, live Azure prices	Strategy + all 5 options	Cheapest option that truly meets targets
Readiness	`gpt-4.1-mini`	7 deterministic checks + read-only tools it calls itself	GO / NO-GO, risks	Any failed check forces NO-GO
Runbook	`gpt-4.1-mini`	Runbook markdown, inventory	Stale / valid per step, corrected runbook	Schema only (advisory)
Planner	`gpt-5-mini`	Strategy, risks, allowlist, safe sequence	21-step drill plan	Allowlisted actions, mandatory sequence, approvals
Validator	`gpt-5-mini`	Failed step, error, events + read-only tools it calls itself	Cause, evidence, fix	Fix allowlisted; never Region A mid-outage; human approves
Reporter	`gpt-4.1-mini`	Measured RTO/RPO, timeline, findings	Summary, remediation items	Numbers rendered by code, not retyped

They investigate, and you can watch. Validator and Readiness call read-only tools themselves (probe, traffic_manager, web_app, sql_failover_group, app_settings_diff, site_recovery); when something is wrong they must call one before answering. Each agent has its own tool allowlist. The live activity panel shows every call: **"AI agent called app_settings_diff() → PAYMENT_PROVIDER only in East Asia"**

Agents decide. The engine acts. Humans approve.

DR touches production, so no model can call an Azure write API: agent tools only read. Every action is allowlisted, ordered, approved and reversible.

Allowlisted actions

Only the engine writes, and only 15 known actions (outage, start app, SQL failover, endpoint on/off, health waits, ASR test). Agent tools only read: probes, settings, status.

Mandatory safe sequence

Every plan must contain heartbeat → outage → SQL failover → reroute → verify → measure → fail back, in that order. Plan safety rules are enforced by the engine.

Human approval gates

A named approver signs off the plan; the engine requires the DB team's 'go' before the outage and the app team's before failback; a human approves any fix. Stale commands are refused.

Kill switch and rollback

One click stops the drill; rollback restores the primary region and the database role.

Fallback for every agent

Foundry agent → model deployment → deterministic rules. The screen shows which one answered.

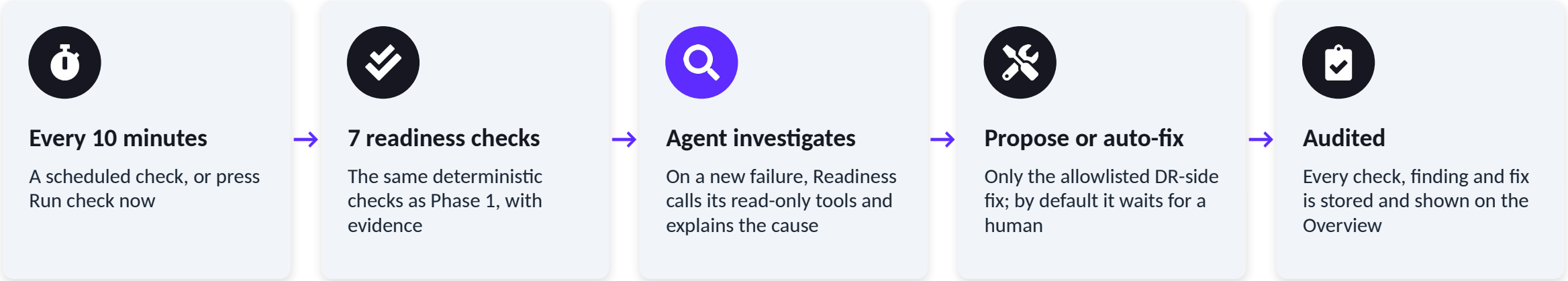
Full audit trail

Every event, approval and measurement is timestamped and stored, then written into the report.

Proven in a chaos drill: we disabled the DR traffic endpoint mid-failover. The Validator diagnosed it from live probes, proposed the allowlisted fix, a human approved, and recovery completed in 57.4 s.

Autopilot: it checks readiness by itself

Every 10 minutes, with no one asking. It never runs during a drill, never starts a drill and never touches traffic.



Three autonomy levels, chosen on the Overview

Observe

Alert only: the finding and the agent's explanation.

Propose fixes (default)

Prepares the allowlisted fix; it waits until a human clicks Approve fix.

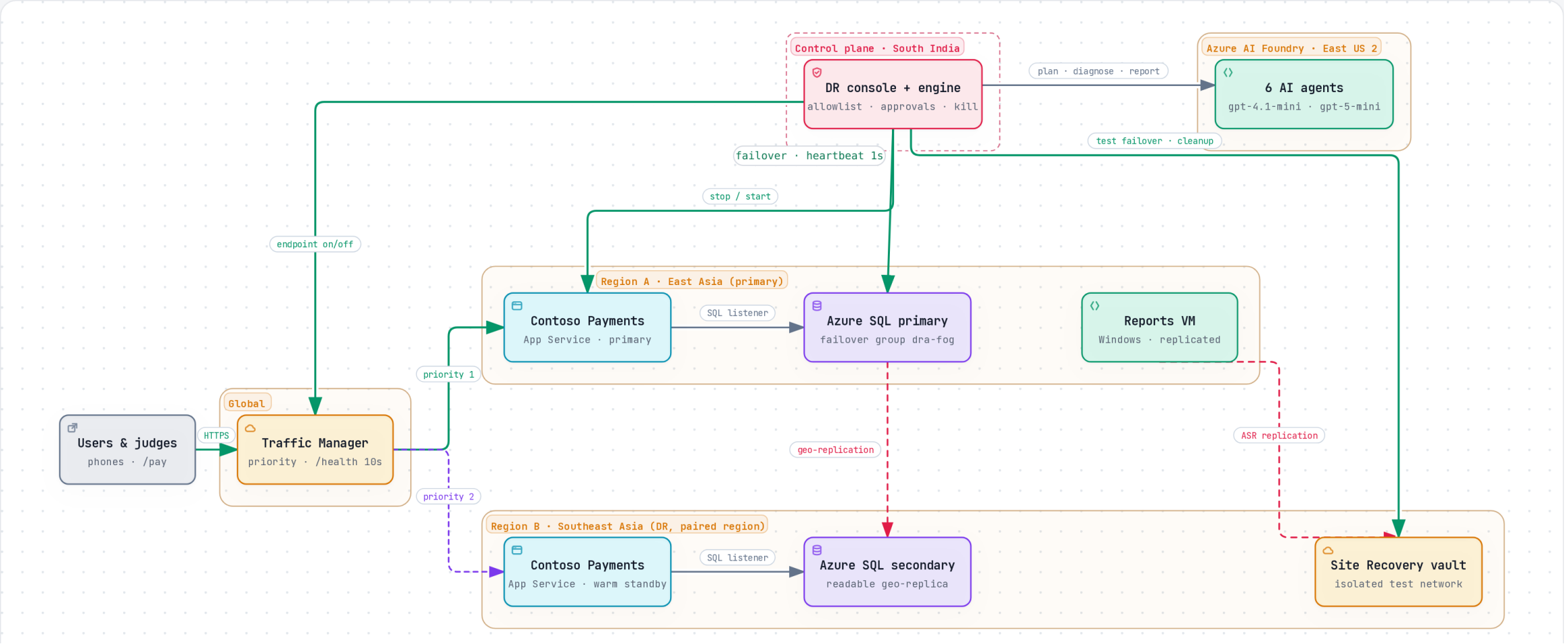
Auto-fix config drift

Applies only the DR-side fix itself: copy a missing app setting to Southeast Asia, audited.

Real event, 19 Oct 2026: a scheduled check, with no one asking, found PAYMENT_PROVIDER missing in Southeast Asia (the DR app answered HTTP 503). Had East Asia failed then, the DR site could not have taken payments.

Architecture: 100% Azure, across an Azure paired region

East Asia (primary) and Southeast Asia (DR). The control plane runs outside both: console in South India, its state and secrets in Central India, agents in East US 2.



App Service ×2 | Azure SQL failover group | Traffic Manager | Site Recovery | AI Foundry Agent Service | Key Vault + managed identity | Storage | Monitor + App Insights | Bicep: one-click up / down

Measured on live Azure, not assumed

Real failovers of the payments service, East Asia → Southeast Asia, 17–19 Oct 2026. Target: RTO 30 min, RPO 15 min.

29.04 s

RTO, 19 Oct drill, no load

27.74–40.43 s across all 10 drills

2.80 s

RPO, proven write by write

same drill: 2 of 45 acknowledged writes lost in the forced failover

1 min 43 s

Failback to primary

1 min 43 s – 2 min 55 s on Premium P0v3 (3 min 44 s on B1); planned

3 min 08 s

VM recovered by Site Recovery

test-failover job; service verified inside the VM at 3 min 55 s

Under payment load (20 writes/s, every write checked), 19 Oct: RTO 27.74 s · RPO 1.95 s (15 of 466 acknowledged writes lost) · failback 2 min 55 s

Where the 29.04 s went (19 Oct drill without load, one clock, segments add up exactly)



Outage injected 0.714 s · SQL forced failover 14.321 s · Traffic Manager reroute 3.782 s · DNS + health checks 9.478 s · Engine bookkeeping 0.745 s

Why judges can trust the numbers: RTO runs from the injected outage to the first healthy check through Traffic Manager from Southeast Asia (checked every second). RPO is proven write by write: every write the database acknowledged until the forced failover completed, including during it, must exist in the new primary. Our earlier "RPO 0" came from a narrower method; our own audit found it.

What it costs to prove recovery

Actual Azure bills and list prices for what we built, compared with the cost of downtime.

$0.09

all AI agent calls

every Foundry agent run across 2 days of building and 9 drills: $0.087 (Azure Cost Management) [12]

$2.44

entire environment so far

two regions, database, VM, console, monitoring: everything, 17–18 Oct 2026 (Azure Cost Management) [12]

$38.26

per month for the PaaS DR pair

2× App Service B1, 2× SQL Basic, Traffic Manager at Azure list prices (Retail Prices API) [13]

A month of this DR set-up costs less than a fifth of a second of downtime at the 2026 average of $15,000 a minute ($38.26 ÷ $250 per second = 0.15 s). [1][13]

Cost-aware by design: the Architect prices every strategy from live Azure list prices and flags anything over budget; the whole environment is created and removed with one command (up.sh / down.sh). For demo week only, App Service Premium P0v3 + SQL S1 (about $6.75 a day), reverted with demo-tier.sh down.

Live demo, and where it goes next

What you will see in the next minutes, on your own phone.

The demo

1. Scan the QR code: Contoso Payments opens on your phone; make a payment
2. Autopilot already found it, unasked: Southeast Asia is missing a setting; the agent shows its tool calls; one approved click fixes it
3. The Planner builds the drill; we approve it and start it under payment load (20 writes/s)
4. The live map shows traffic and the primary database moving; your phone reconnects to Southeast Asia in about 30–40 s
5. Failback, then the report: RTO, RPO write by write, compared with earlier drills

Roadmap (not in the demo)

Scheduled drills

on a set cadence (e.g. quarterly, as SEBI asks) and after every infrastructure change

More services

AKS, Cosmos DB, Storage, Service Bus as new allowlisted actions

Azure Chaos Studio

standard fault library alongside our own faults

Approvals where teams work

Microsoft Teams approval cards

Portfolio view

resilience score across many applications

Sources

Every figure in this deck is taken from the source below. Our own figures come from live drills on Azure.

[1] Splunk with Oxford Economics, The Hidden Costs of Downtime 2026, 19 May 2026 (2,000 Global 2000 executives). newsroom.cisco.com (The $600 Billion Wake-up Call)

[2] Uptime Institute, Annual Outage Analysis 2026, press release 13 May 2026. uptimeinstitute.com

[3] Parametrix estimate of losses from the AWS us-east-1 outage of 20 Oct 2025 (Insurance Edge, 30 Oct 2025).

[4] RBI order to HDFC Bank, 2 Dec 2020 (Outlook Business, 3 Dec 2020); curbs lifted 17 Aug 2021 (cards) and 11 Mar 2022 (digital) (Business Standard, 12 Mar 2022).

[5] NSE and NSE Clearing settle the 24 Feb 2021 trading-halt case with SEBI for ₹72.64 crore (Moneylife; Business Standard, Jun 2023); ≈ $8.8M at June 2023 rates of ₹82–83 per US$.

[6] RBI action on Kotak Mahindra Bank under Sec. 35A, 24 Apr 2024 (TechCrunch); restrictions lifted 12 Feb 2025 (Business Standard).

[7] RBI, Cybersecurity, Technology: Risk, Resilience and Assurance Framework Directions, 31 Jul 2026, summarised by KPMG India, Sep 2026.

[8] SEBI Cybersecurity and Cyber Resilience Framework: RTO 2 hours, RPO 15 minutes (clarification reported by PTI / The Week, 28 Aug 2025).

[9] SEBI framework for market infrastructure institutions: DR site within 45 minutes, quarterly drills (Business Standard, 22 Mar 2021).

[10] Regulation (EU) 2022/2554 (DORA), Articles 11(6), 12, 35(7)–(8), 50, 51. eur-lex.europa.eu/eli/reg/2022/2554/oj

[11] Cockroach Labs with Wakefield Research, The State of Resilience 2025 (1,000 executives, Aug–Sep 2024). cockroachlabs.com

[12] Azure Cost Management, actual cost by service in US$ (CostUSD), month to date, queried 18 Oct 2026 (our environment).

[13] Azure Retail Prices API (prices.azure.com), East Asia / Southeast Asia list prices for App Service B1 Linux, SQL Database Basic, Traffic Manager, 18 Oct 2026.

Our measurements: 10 completed drills recorded by the DR Agent (17–19 Oct 2026), including drills under payment load; each report keeps the raw timestamps, events, writes and approvals.



Don't assume you can recover.

Let the DR Agent test it for real, and prove it, whenever you need.

Real DR drills
10 live drills, RTO 27.7–40.4 s, data loss proven write by write, also under payment load

Autonomous, not reckless
Autopilot checks readiness by itself; agents investigate with read-only tools

Built 100% on Azure
AI Foundry, SQL failover groups, Traffic Manager, Site Recovery, Monitor

Safe by design
Allowlisted actions, human approval, kill switch, full audit trail

Thank you · Team Azure Armada